(ii) Sum of two numbers is 45

This statement can be written as $x + y = 45$.

This equation $x + y = 45$ is formed by two variables x and y whose highest power is 1. Hence, we call it as a linear equation in two variables.

Now, in this class we shall learn to solve linear equations in one variable only. You will learn to solve other type of equations in higher classes.



Note

The equations so formed with power more than 1 of its variables, (2,3....etc.) are called as quadratic, cubic equations and so on.

Examples: (i) $x^2 + 4x + 7 = 0$ is a quadratic equation.

(ii) $5x^3 - x^2 + 3x = 10$ is a cubic equation.



Try these

Identify which among the following are linear equations.

(i) $2 + x = 19$ (ii) $7x^2 - 5 = 3$ (iii) $4p^3 = 12$ (iv) $6m+2$ (v) $n=10$

(vi) $7k - 12 = 0$ (vii) $\frac{6x}{8} + y = 1$ (viii) $5 + y = 3x$ (ix) $10p+2q=3$ (x) x^2-2x-4

Convert the following statements into linear equations:

Example 3.28

7 is added to a given number to give 19.

Solution:

Let the number be n.

When 7 is added to this number we get $n + 7$.

This result is to give 19.

Therefore, the equation is $n + 7 = 19$.

Example 3.29

The sum of 4 times a number and 18 is 28.

Solution:

Let the number be x.

4 times the number is $4x$.

Adding 18 now, we get $18 + 4x$.

Now the result should be 28.

Thus, the equation has to be $18 + 4x = 28$.



Think

(i) Is $t(t-5)=10$ a linear equation? Why?

(ii) Is $x^2 = 2x$, a linear equation? Why?

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